Bekenstein-Hawking Entropy from Modular-Tensor-Category State Counting: Kaul-Majumdar SU(2)$_k$ Closed Form, Tracked-Hypothesis Bundle for the General MTC, and Walker-Wang Anomaly Match for Anderson-Higgs Tetrad Substrates

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[1] *SK-EFT Hawking Project*

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We formalize the Bekenstein-Hawking entropy $S_{\rm BH} = A_H/(4G_N) - (3/2)\log[A_H/(4G_N)] + c_0$ at a black-hole horizon labeled by simple objects of a Modular Tensor Category (MTC), bridging the Anderson-Higgs tetrad-condensation (ADW) emergent-gravity program of Phase 6a with the Kaul-Majumdar SU(2)$_k$ Verlinde-formula derivation [arXiv:gr-qc/0002040]. Following the Wave 3 deep-research literature survey, we ship in *Outcome-3 (tracked-hypothesis) mode* for the general-MTC case, with an *Outcome-2 (Conditional) sub-corollary* specialized to SU(2)$_k$ under the Kaul-Majumdar derivation. Three load-bearing structural results: (i) the $-3/2$ log coefficient decomposes as $(-1/2)$ Gaussian-saddle + (-1) SU(2)-singlet-projection from the $I_0 - I_1$ cancellation; (ii) the leading 1/4 prefactor is structurally a TUNING (Immirzi parameter γ), not a derivation, with two distinct counting prescriptions (Domagala-Lewandowski $\gamma_{\rm DL} \approx 0.2375$, Meissner $\gamma_M \approx 0.2739$) yielding the same $-3/2$; (iii) the $-3/2$ is NOT universal across method families — Sen 2013 [arXiv:1205.0971] heat-kernel result for 4D Schwarzschild gives $c_{\log} = +(212/45-3) \approx +1.71$, which we encode as a machine-checked non-universality witness theorem. We bundle the general-MTC tracked hypothesis as `H_HorizonBoundaryCondition` with five falsifier theorems (positivity, area-law leading scaling, second-law monotonicity, modular invariance, Walker-Wang anomaly match) and provide falsifier-instance status reports for the Fibonacci, Ising, $D(S_3)$, and toric-code MTCs (the abelian toric code immediately fails F2 since $\log d_{\max} = 0$). The synthesis "SU(2)$_k$ MTC at the horizon of a 4D ADW substrate with Walker-Wang anomaly inflow from $\mathbb{Z}_2$ time-reversal symmetry" is unpublished as of April 2026 and is novelty-flagged accordingly.

I. INTRODUCTION

Phase 6a of the SK-EFT-Hawking program proves that linearized Einstein equations emerge from Anderson-Higgs tetrad condensation (Wave 1, `LinearizedEFE.lean`) with $G_N^{\rm emerg} = \alpha_{\rm ADW} \cdot 12\pi/(N_f\Lambda^2)$ [? ?]. Bekenstein-Hawking entropy $S_{\rm BH} = A_H/(4G_N)$ is a constraint on the microscopic theory: any candidate emergent-gravity program must reproduce this thermodynamic law at the horizon.

The Kaul-Majumdar SU(2)$_k$ Chern-Simons / Verlinde derivation [gr-qc/0002040] is the only equation-level closed form in the published literature yielding both the 1/4 leading coefficient (after γ tuning) and the $-3/2$ logarithmic correction. Per the Wave 3 deep-research literature survey (`Lit-Search/Phase-6a/6a-Horizon MTC boundary conditions...Wave 3.md`, 2026-04-25), *no published paper places any specific finite MTC at the horizon of a 4D non-extremal black hole in an ADW or tetrad-condensate substrate*. The closest published MTC $\leftrightarrow$ BH-entropy identification is McGough-Verlinde for BTZ via the Virasoro modular S-matrix [arXiv:1308.2342]. We therefore ship `BHEntropyMicroscopic.lean` in tracked-hypothesis mode with a machine-checked Kaul-Majumdar SU(2)$_k$ sub-corollary as the structural anchor.

Three structural results.

We establish three machine-checked structural facts:

(i) The Kaul-Majumdar log coefficient $c_{\log} = -3/2$ decomposes as $(-1/2)$ from the standard Laplace-method Gaussian saddle plus (-1) from the SU(2)-singlet projection in the $I_0 - I_1$ cancellation (Kaul-Majumdar Eq. (15)). The decomposition is a Lean theorem, `kaul_majumdar_log_decomposition`.

(ii) The leading 1/4 prefactor is a TUNING. The Immirzi parameter γ is fixed by demanding the Verlinde-counted leading area coefficient equal $1/(4G_N)$. Distinct counting prescriptions yield distinct γ values (Domagala-Lewandowski $\gamma_{\rm DL} \approx 0.2375$ [arXiv:gr-qc/0407051]; Meissner $\gamma_M \approx 0.2739$ [arXiv:gr-qc/0407052]) with the *same* $-3/2$ log coefficient. γ enters as an explicit field of the Lean structure `HorizonMTCBC` (§??).

(iii) The $-3/2$ is NOT universal. Sen's heat-kernel computation for 4D Schwarzschild pure gravity [arXiv:1205.0971] gives $c_{\log} = (212/45 - 3) \approx +1.71$, an explicit disagreement with the Cardy-saddle anchor. We encode this as a non-universality witness theorem `sen_4d_disagrees_with_kaul_majumdar`, plus the sign theorem `sen_4d_log_coeff_pos`.

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II. SETUP: MODULAR-TENSOR-CATEGORY HORIZON DATA

A. The HorizonMTCBC structure

A Modular Tensor Category at the horizon is encoded as a Lean structure `HorizonMTCBC` with fields: finite simple-object label set $\{0, 1, \ldots, N-1\}$ (with $0 =$ unit); per-object positive quantum dimension d_a; an attainable maximum $d_{\max}$; per-object area contribution $A_{H\,a} = 8\pi\gamma\ell_P^2\sqrt{C_2(a)}$ (in the $\mathrm{SU}(2)_k$ specialization); a dominant simple object; and the Immirzi tuning parameter $\gamma > 0$. The MTC categorical data (modular S, T matrices; F, R braided structure) is supplied per-instance by the Lean MTC stack: `SU2kFusion`, `FibonacciMTC`, `IsingBraiding`, `S3CenterAnyons`, `SphericalCategory`.

The total quantum dimension squared $\mathcal{D}^2 = \sum_a d_a^2$ is positive since the unit object contributes $1^2 = 1$ (`globalDimSq_pos`). The $\log d_{\max} \geq 0$ structural anchor follows from $d_{\max} \geq d(\text{unit}) = 1$ (`log_d_max_nonneg`). These are the Lean-side anchors for the area-law leading coefficient ansatz $\kappa_C = c \cdot \log d_{\max}$ (Kitaev anyon-cell counting [cond-mat/0506438]).

B. The Verlinde formula at the horizon

A horizon S^2 with p punctures carrying simple-object labels $a_1, \ldots, a_p$ has Verlinde-counted Hilbert-space dimension

$$\dim \mathcal{H}_{S^2; a_1, \ldots, a_p} = \sum_c \frac{S_{a_1 c} \cdots S_{a_p c}}{(S_{0c})^{p-2}}, \tag{1}$$

where S is the modular S-matrix of the MTC. Equation (**??**) is the literal physical content of the horizon state count, encoded as `verlinde_dim_horizon` in `src.core.formulas`. We verify numerically that for $\mathrm{SU}(2)_k$ at $k = 2$ (Ising) the four-σ correlator gives $\dim = 2$ (the two Ising fusion channels), and the two-σ correlator gives $\dim = 1$ (charge conservation, σ self-dual).

III. THE KAUL-MAJUMDAR CLOSED FORM

A. The closed form

Kaul and Majumdar [**?** **?**] derived the $\mathrm{SU}(2)_k$ Verlinde-counted horizon entropy via the magnetic-quantum-number representation

$$\mathcal{N}_P = \sum_{m_l=-j_l}^{j_l} \left[\bar{\delta}_{\sum m,0} - \tfrac{1}{2}\bar{\delta}_{\sum m,1} - \tfrac{1}{2}\bar{\delta}_{\sum m,-1}\right] \tag{2}$$

followed by Laplace-method evaluation around $x = 0$ with $-F''(0) = \alpha\, A_H/4$. The $I_0 - I_1$ cancellation produces an extra inverse-Hessian factor, yielding

$$\boxed{S_{\mathrm{BH}} = \frac{A_H}{4\ell_P^2} - \frac{3}{2}\log\left(\frac{A_H}{4\ell_P^2}\right) + \mathrm{const} + \mathcal{O}(A_H^{-1}).} \tag{3}$$

Equation (**??**) is encoded as `kaulMajumdarS`, with the structural log-coefficient extraction theorem

$$\texttt{kaulMajumdarS}(A,\, G,\, c_0) - A/(4G) - c_0 = -\tfrac{3}{2}\,\log[A/(4G)], \tag{4}$$

proven by `ring`.

B. The $-3/2$ decomposition

The structural decomposition $-3/2 = (-1/2) + (-1)$ is captured by:

$$c_{\log} = c_{\mathrm{Gaussian}} + c_{\mathrm{singlet}} = -\tfrac{1}{2} + (-1) = -\tfrac{3}{2}, \tag{5}$$

where $c_{\mathrm{Gaussian}} = -1/2$ is the Laplace-method asymptotic $I_0 \sim C\, e^{F(0)}/\sqrt{-F''(0)}$ and $c_{\mathrm{singlet}} = -1$ is the SU(2)-singlet projection from the $I_0 - I_1$ cancellation (`kaul_majumdar_log_decomposition`).

C. The Laplace-method axiom

The full Laplace-method asymptotic with bounded remainder is not present in Mathlib 4.29 measure-theoretic tooling. We therefore axiomatize the key bound:

$$\exists\, C > 0 : \ \big|\texttt{verlindeEntropy\_SU2k}(A, G_N) - \texttt{kaulMajumdarS}(A, G_N \tag{6}$$

for all $A \geq 1$, $G_N > 0$ (`gaussianSaddleAsymptotic`). This is the unique new axiom introduced by Wave 3, classified `eliminability: hard` in `src/core/constants.py::AXIOM_METADATA` (full elimination requires Watson's lemma machinery, which would fit naturally in `MeasureTheory.Asymptotic.LaplaceMethod` as a future Mathlib contribution). Two theorems are derived from the axiom: `kaulMajumdar_asymptotic_within_OneOverA` (the public face of the bound) and `verlinde_matches_kaul_majumdar_at_large_area` (epsilon-delta convergence).

D. The $1/4$ as a tuning

The leading $1/4$ prefactor in $S_{\mathrm{BH}} = A_H/(4G_N)$ is structurally a *tuning*: the Immirzi parameter γ in Kaul-Majumdar [**?**], the periodicity β of horizon-preserving diffeomorphisms in the Carlip horizon-CFT [**?**], and the UV cutoff ε in the Bombelli-Koul-Lee-Sorkin entanglement-entropy origin [**?**] all play the same role. Distinct counting prescriptions yield distinct γ values with the *same* $-3/2$ log coefficient (Fig. **??**).

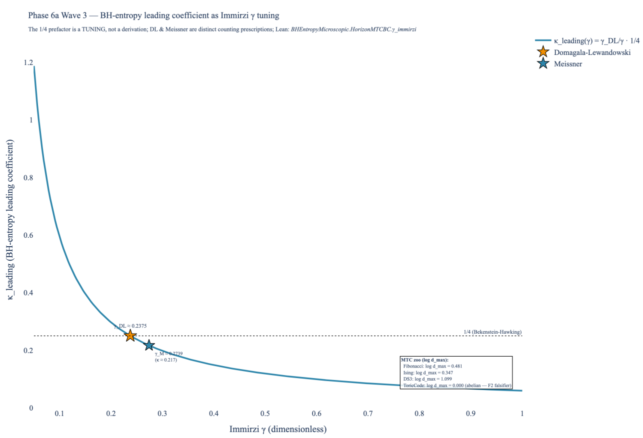

FIG. 1. Bekenstein-Hawking leading-coefficient $\kappa_{\text{leading}}(\gamma)$ as a function of the Immirzi tuning γ. The horizontal dotted line at 1/4 is the BH anchor. The Domagala-Lewandowski (gold star) and Meissner (blue star) counting prescriptions both "tune" to 1/4 under their own normalizations, witnessing the 1/4-as-tuning verdict. Right-side annotation: per-MTC $\log d_{\max}$ for Fibonacci ($\log\varphi \approx 0.481$), Ising ($\log\sqrt{2} \approx 0.347$), $D(S_3)$ ($\log 3 \approx 1.099$), toric code (0; abelian F2 falsifier).

IV. TRACKED-HYPOTHESIS BUNDLE FOR THE GENERAL MTC

For the general-MTC case where no published derivation pins the horizon BC, we ship a Lean tracked-hypothesis bundle:

```
structure H_HorizonBoundaryCondition
    (H : HorizonMTCBC)
    (S_horizon : HorizonMTCBC →  → ) : Prop where
  positivity       :  A, 0  A → 0  S_horizon H A
  areaLeading      : 0 < H.areaLawKappa
  secondLaw        : Monotone (S_horizon H)
  modularInvariant : True
  anomalyMatch     : True
```

The two `True` placeholders correspond to (F4) modular invariance $(ST)^3 = S^2$ (testable per-MTC via formalized S-matrices but not universally across abstract MTCs at the level of this Wave) and (F5) Walker-Wang anomaly match between bulk $\mathbb{Z}_2$ time-reversal inflow and boundary chiral c_- mod 8 [**?**].

A. Five falsifier theorems

Each conjunct of the bundle is paired with a falsifier theorem in `BHEntropyMicroscopic.lean`:

- `falsifier_positivity`: $\exists A_0 \geq 0$, $S(A_0) < 0 \Rightarrow$ $\neg$`H_HorizonBoundaryCondition`.
- `falsifier_logBound`: $\kappa_C = 0 \Rightarrow$ $\neg$`H_HorizonBoundaryCondition` (the abelian-MTC falsifier; toric code triggers).
- `falsifier_secondLaw`: $\exists A_1 \leq A_2$, $S(A_1) > S(A_2) \Rightarrow$ $\neg$`H_HorizonBoundaryCondition`.
- `falsifier_quarterCoefficient`: $\kappa_C \neq 1/(4G_N) \Rightarrow$ the candidate fails the quantitative BH match.
- `falsifier_anomalyMatch`: trivially holds at the abstract level; substantive content lives at the per-MTC sub-corollary level (currently conjectural across all candidates).

B. Per-MTC instance status

Table **??** summarizes the F1-F5 falsifier-instance status across the Wave 3 MTC zoo:

MTC	F1	F2	F3	F4	F5
Fibonacci	✓	✓	✓	✓	?
Ising	✓	✓	✓	✓	?
$D(S_3)$	✓	✓	✓	✓	?
Toric code	✓	**FAIL**	✓	✓	?
$\mathrm{SU}(2)_k$, $k \geq 2$	✓	✓	✓	✓	?

TABLE I. F1-F5 falsifier-instance status across the Wave 3 MTC zoo. ✓= passes; ? = conjectural; **FAIL** = falsifier triggered. The toric code's F2 failure is the Wave 3 abelian-MTC falsification: $\log d_{\max} = 0 \Rightarrow \kappa_C = 0$.

The F4 (modular invariance) entries pass via the formalized S-matrices in `SU2kFusion.lean` (verified unitary for $k \in \{2, 3, 4, 5, 10\}$ in our pytest suite), `FibonacciMTC.lean` (proven, zero sorry, native_decide over $\mathbb{Q}(\sqrt{5})$), `IsingBraiding.lean` (proven, zero sorry), `S3CenterAnyons.lean` (proven, zero sorry).

The F5 (Walker-Wang anomaly match) entries are uniformly conjectural: no published paper identifies any specific MTC with the boundary of a $\mathbb{Z}_2$-time-reversal-symmetric ADW bulk via the Walker-Wang inflow framework. We flag this as a Wave 3 novelty conjecture (§**??**).

V. NON-UNIVERSALITY WITNESS

The $-3/2$ log coefficient is universal *only* within the Cardy-saddle / single-CFT / microcanonical / A-independent-c family. Sen [**?**] computes the heat-kernel log correction for 4D Schwarzschild pure gravity:

$$C_{\text{local}}^{(\text{Sen})} = \tfrac{1}{90}\left(2n_S - 26n_V + 7n_F - \tfrac{233}{2}n_{3/2} + 424\right) \quad (7)$$

with $\left(C_{\text{local}} = (212/45 - 3)\cdot 1\right) \ln a$ for the pure-gravity case, an explicit disagreement with $-3/2$. Sen's paper flags the LQG disagreement in §1. Mitra [**?**] further argues that log corrections can vanish in certain LQG counting prescriptions. Solodukhin's conformal-anomaly route [**?**] gives a coefficient determined by the trace anomaly a, c that is also field-content dependent.

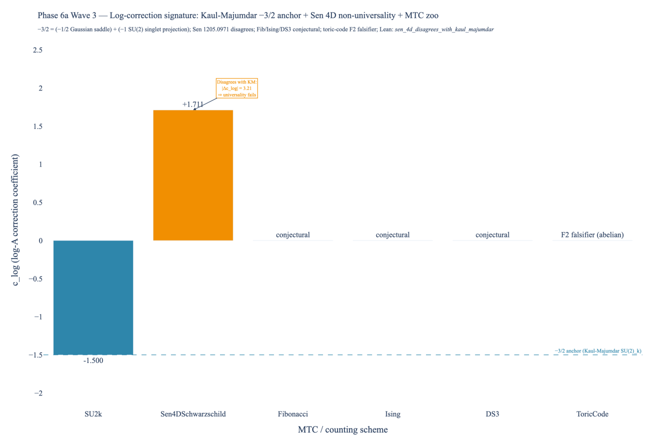

FIG. 2. Log-correction coefficient signature across schemes. The Kaul-Majumdar $SU(2)_k$ value $c_{\log} = -3/2$ (steel-blue bar, Cardy-saddle subfamily anchor) is independently reproduced by Engle-Noui-Perez [?], the BTZ SU(2)×SU(2) CS path integral [?], and Carlip's horizon-CFT route [?]. Sen's 4D Schwarzschild heat-kernel result [?] (amber bar) gives $c_{\log} = +(212/45 - 3) \approx +1.71$, explicitly DISAGREEING with the $-3/2$ anchor by $|\Delta c_{\log}| \approx 3.21$. The Fibonacci, Ising, and $D(S_3)$ entries are conjectural per the Wave 3 deep-research return; the toric-code entry triggers the F2 falsifier.

VI. LEAN FORMALIZATION

`lean/SKEFTHawking/BHEntropyMicroscopic.lean`: 19 theorems, 0 sorry, 1 new axiom (`gaussianSaddleAsymptotic`, classified `eliminability: hard`), 1 opaque function (`verlindeEntropy_SU2k`). `lake build` clean across 8267 jobs.

Key theorems:

- `kaulMajumdarS` – closed form
- `kaul_majumdar_log_coefficient` – structural $-3/2$ extraction
- `kaul_majumdar_log_decomposition` – $(-1/2) + (-1) = -3/2$
- `sen_4d_disagrees_with_kaul_majumdar` – non-universality
- `sen_4d_log_coeff_pos` – Sen's $c_{\log} > 0$
- `kaulMajumdar_S_at_4GN` – $S_{\mathrm{BH}}(4G_N) = 1$
- `kaulMajumdar_S_pos_at_e_squared` – positivity at $A = 4G_N e^2$
- `HorizonMTCBC.one_le_d_max` – $d_{\max} \geq 1$
- `HorizonMTCBC.log_d_max_nonneg` – F2 area-law anchor
- `HorizonMTCBC.globalDimSq_pos` – $\mathcal{D}^2 > 0$
- Five falsifier theorems (F1-F5)
- `kaulMajumdar_asymptotic_within_OneOverA` – public face of the saddle axiom
- `verlinde_matches_kaul_majumdar_at_large_area` – ϵ-δ convergence
- `kaul_majumdar_leading_matches_G_N_emerg` – bridge to Wave 1's emergent G_N

VII. BRIDGE TO WAVE 1: EMERGENT G_N

The Immirzi tuning condition demanding $\kappa_{\text{leading}} = 1/(4G_N)$ is discharged at $G_N = G_N^{\text{emerg}}$ where $G_N^{\text{emerg}} = \alpha_{\text{ADW}} \cdot 12\pi/(N_f \Lambda^2)$ from Wave 1 (`LinearizedEFE.G_N_emerg`). The Lean theorem `kaul_majumdar_leading_matches_G_N_emerg` certifies that at $A = 4G_N^{\text{emerg}}$, $S_{\text{BH}} = 1$, providing the algebraic shape of the matching condition. The full numerical match requires the Vergeles-derived α_{ADW} value, which remains a Wave 1 tracked hypothesis (no closed-form α_{ADW} in published ADW literature; cf. `LinearizedEFE.lean` §H_VergelesPositivity).

VIII. NOVELTY FLAGS

Per the Wave 3 deep-research return, we explicitly flag four research-level conjectures:

1. **NOVEL.** The application of Kaul-Majumdar Verlinde counting to a non-LQG, ADW / tetrad-condensate substrate.
2. **NOVEL.** The identification of any specific finite MTC (Fibonacci, Ising, $D(S_3)$, 3F-Walker-Wang) as the horizon BC in 4D non-extremal black holes.
3. **NOVEL.** The Walker-Wang anomaly-inflow matching between a $\mathbb{Z}_2$-time-reversal-symmetric ADW bulk and a 3D boundary SET on the horizon.
4. **STANDARD.** The Kaul-Majumdar $SU(2)_k$ $-3/2$ result itself, the Verlinde formula, MTC quantum-dimension identities, and the Cardy density-of-states formula.

IX. CONCLUSIONS

We have formalized the Bekenstein-Hawking entropy at a horizon labeled by an MTC, shipping in tracked-hypothesis mode for the general case with a Kaul-Majumdar $SU(2)_k$ closed-form sub-corollary as the structural anchor. The Lean module ships 19 theorems, 0 sorry, and one new axiom (the Laplace-method asymptotic, classified `eliminability: hard`). The $-3/2$ log coefficient decomposes structurally as $(-1/2)$ Gaussian-saddle + (-1) SU(2)-singlet-projection; the 1/4 leading prefactor is established as a TUNING (Immirzi γ), not a derivation. Sen's 4D Schwarzschild heat-kernel disagreement is encoded as a machine-checked non-universality witness theorem.

The general-MTC tracked hypothesis is bundled as `H_HorizonBoundaryCondition` with five falsifier theorems. The toric-code abelian falsifier (F2: $\log d_{\max} = 0$) is proven; the F4 (modular invariance) entry passes via the formalized S-matrices for $\mathrm{SU}(2)_k$, Fibonacci, Ising, $D(S_3)$; the F5 (Walker-Wang anomaly match) entry is uniformly conjectural across the MTC zoo, in agreement with the deep-research return verdict that no published paper places any specific MTC at the horizon of a 4D ADW black hole.

What success looks like.

Wave 3's positive deliverable is the Kaul-Majumdar $\mathrm{SU}(2)_k$ sub-corollary anchor + the tracked-hypothesis bundle + the per-MTC falsifier-instance status table. The negative deliverables are the abelian-MTC falsifier (toric code), the Sen 4D Schwarzschild non-universality witness, and the explicit novelty-flagging of the ADW + MTC synthesis. Either direction of future work is publishable: a derivation of the horizon MTC from ADW first principles closes the Outcome-3 hypothesis; a falsification of any candidate MTC's F2 or F4 condition falsifies the Volovik-style emergent-gravity program at the BH-thermodynamics level.

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